

PNP Log-Concentration Transition

Side-by-Side Comparison with the Concentration Formulation

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1 Purpose and scope

The active production stack (`Forward/bv_solver/forms_logc.py`) uses $u_i = \ln c_i$ as primary unknowns instead of c_i . This note maps every equation in `docs/PNP Equation Formulations.tex` to its log-concentration counterpart so the transition can be verified line by line. The mathematical change is a single substitution; the practical change is the elimination of negative-concentration Newton iterates and the resulting floor/regularization terms. The Poisson equation is structurally unchanged modulo $c_i \rightarrow e^{u_i}$ in its source.

Throughout, the substitution

$$c_i = e^{u_i}, \quad \nabla c_i = e^{u_i} \nabla u_i = c_i \nabla u_i,$$

is applied to every term in the strong and weak forms. The combination $\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi$ that appears in the Nernst–Planck flux factors as $c_i (\nabla u_i + \frac{z_i F}{RT} \nabla \phi)$, which is the central simplification.

2 Strong form

	Concentration (c_i)	Log-concentration ($u_i = \ln c_i$)
NP equation	$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(D_i \left[\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi \right] \right)$	$\frac{\partial e^{u_i}}{\partial t} = \nabla \cdot \left(D_i e^{u_i} \left[\nabla u_i + \frac{z_i F}{RT} \nabla \phi \right] \right)$
NP flux	$\mathbf{J}_i = -D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi \right)$	$\mathbf{J}_i = -D_i e^{u_i} \left(\nabla u_i + \frac{z_i F}{RT} \nabla \phi \right)$
Poisson	$-\nabla \cdot (\epsilon \nabla \phi) = \sum_i z_i F c_i$	$-\nabla \cdot (\epsilon \nabla \phi) = \sum_i z_i F e^{u_i}$
BV flux BC on Γ_{elec}	$-D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi \right) \cdot \mathbf{n} = \pm \frac{j}{n_e F}$	$-D_i e^{u_i} \left(\nabla u_i + \frac{z_i F}{RT} \nabla \phi \right) \cdot \mathbf{n} = \pm \frac{j}{n_e F}$
Bulk concentration BC on Γ_{conc}	$c_i = c_{0,i}$	$u_i = \ln c_{0,i} \quad (\text{with floor } \max(c_{0,i}, 10^{-20}) \text{ for product species with } c_{0,i} = 0)$

What is unchanged: the Poisson PDE structure, the dielectric ϵ , the valences z_i , E_{eq} , the bulk Poisson source sum, and the choice of which boundary carries the BV flux. Only the representation of c_i has changed.

What is new: the Dirichlet bulk BC is now $u_i = \ln c_{0,i}$, which is undefined for $c_{0,i} = 0$. The code uses a floor $c_{0,i} \leftarrow \max(c_{0,i}, 10^{-20})$ for product species (`forms_logc.py:441,510`); this matters for H_2O_2 , whose physical bulk IC is zero.

3 Weak form

Multiplying the NP equation by $v \in V$ and integrating by parts (with the electrode boundary handled by the BV flux condition):

	Concentration	Log-concentration
NP weak form	$\int_{\Omega} \frac{\partial c_i}{\partial t} v \, dx + \int_{\Omega} D_i [\nabla c_i + \frac{z_i F}{RT} c_i \nabla \phi] \cdot \nabla v \, dx$ $= \int_{\Gamma_{\text{elec}}} \left(\mp \frac{j}{n_e F} \right) v \, ds$	$\int_{\Omega} \frac{\partial e^{u_i}}{\partial t} v \, dx + \int_{\Omega} D_i e^{u_i} [\nabla u_i + \frac{z_i F}{RT} \nabla \phi] \cdot \nabla v \, dx$ $= \int_{\Gamma_{\text{elec}}} \left(\mp \frac{j}{n_e F} \right) v \, ds$
Poisson weak form	$\int_{\Omega} \epsilon \nabla \phi \cdot \nabla w \, dx = \int_{\Omega} \left(\sum_i z_i F c_i \right) w \, dx$	$\int_{\Omega} \epsilon \nabla \phi \cdot \nabla w \, dx = \int_{\Omega} \left(\sum_i z_i F e^{u_i} \right) w \, dx$

3.1 Time discretization (backward Euler)

The code uses backward Euler in c -space, even though the unknown is u :

$$\boxed{\int_{\Omega} \frac{e^{u_i^{n+1}} - e^{u_i^n}}{\Delta t} v \, dx \neq \int_{\Omega} e^{u_i^{n+1}} \frac{u_i^{n+1} - u_i^n}{\Delta t} v \, dx}$$

The left-hand form (used in `forms_logc.py:294`) preserves discrete species mass directly; the right-hand form would not. This is the only point in the discretization where the choice of variable could in principle drift from the concentration formulation, and the code is correct.

4 Butler–Volmer boundary residual

The BV reaction rate is

$$R_{\text{BV}} = k_0 \left[c_{\text{cat}} e^{-\alpha n_e \eta / V_T} - c_{\text{anod}} e^{(1-\alpha) n_e \eta / V_T} \right], \quad \eta = \phi_m - \phi_s - E_{\text{eq}}.$$

Both branches in `forms_logc.py` evaluate the same rate mathematically; they differ in how they assemble it numerically.

4.1 Standard branch (`bv_log_rate=False`)

Evaluates surface concentrations directly from the (clamped) log unknowns:

$$c_{\text{surf},i} = \exp(\text{clamp}(u_i, \pm 30)), \quad R = k_0 c_{\text{surf,cat}} e^{-\alpha n_e \eta / V_T} - k_0 c_{\text{surf,anod}} e^{(1-\alpha) n_e \eta / V_T}.$$

This is the literal port of the original BV formulation to the log-c unknowns. The clamp ± 30 is required because u can wander wildly during Newton iteration and `exp(u)` would otherwise overflow.

4.2 Log-rate branch (bv_log_rate=True, current default)

Algebraically rearranges the cathodic and anodic terms so the only exp applies to a single scalar sum, deferring evaluation:

$$\begin{aligned}\log r_{\text{cath}} &= \ln k_0 + u_{\text{cat}} + \sum_j p_j (u_j - \ln c_{\text{ref},j}) - \alpha n_e \eta / V_T \\ \log r_{\text{anod}} &= \ln k_0 + u_{\text{anod}} + (1 - \alpha) n_e \eta / V_T \\ R &= e^{\log r_{\text{cath}}} - e^{\log r_{\text{anod}}}.\end{aligned}$$

The u_i enter *unclamped*. There is no c_{surf} floor in this branch. This is what eliminates the phantom R₂ sink at $V_{\text{RHE}} > +0.30$ V (when $u_{\text{H}_2\text{O}_2}$ approaches the -30 floor of the standard branch).

The two branches are identical in exact arithmetic; the difference is purely numerical. See docs/clipping-conventions.md for the full discussion.

5 Active clamps in the production solver

The transition to log-c does *not* eliminate clipping. Three distinct guards exist in the code; only one was removed by the log-rate algebraic rearrangement.

Clip / clamp	Where in code	Active?	Guards against
η -clip ± 50 on η_{scaled}	_build_eta_clipped (l.226)	yes	overflow of $\exp(\alpha n_e \eta)$
u -clamp ± 30 on bulk u_i	bulk PDE residual (l.203)	yes	overflow / underflow of e^{u_i} as PDE coefficient
c_{surf} clamp inside BV	legacy BV branch only (l.369–389)	no (log-rate)	under-floored e^u corrupting BV at high V

Why the bulk u -clamp cannot be eliminated by the log-rate trick: in the BV residual the entire reaction rate collapses to a single $\exp(\text{scalar sum})$, so the exp can be deferred. In the bulk PDE, e^{u_i} appears as a *coefficient* multiplying ∇v and other terms that vary across the domain — there is no single exp to defer. The u -clamp is therefore floating-point safety only; widen to $u_{\text{max}} = 100$ when $V_{\text{RHE}} > +0.30$ V because $u_{\text{H}_2\text{O}_2}$ at SS itself approaches -30 in that regime.

Why the η -clip cannot be eliminated either: $\exp(\pm 138) \sim 10^{\pm 60}$ overflows double precision; η -clip at ± 50 is the standard guard. R₂ ($E_{\text{eq}} = 1.78$ V, $n_e = 2$) unclips when $|V - E_{\text{eq}}| < 50 V_T = 1.285$ V, i.e. $V > +0.495$ V. R₁ ($E_{\text{eq}} = 0.68$ V, $n_e = 1$) unclips for $V > -0.605$ V, so it is essentially always unclipped on the production grid.

6 Verification checklist

Items to spot-check while reading the code against this writeup. All line numbers refer to Forward/bv_solver/forms_logc.py unless otherwise noted.

1. Flux factoring (l. 291).

Jflux = D[i] * c_i * (grad(u_i) + grad(drift))

Confirms $\mathbf{J} = -D c (\nabla u + \frac{zF}{RT} \nabla \phi)$.

2. **Time derivative** (l. 294).

`((c_i - c_old) / dt_const) * v * dx`
Confirms $\Delta(e^u)/\Delta t$, not $e^u \Delta u/\Delta t$.

3. **Poisson source** (l. 428).

`sum(z[i] * ci[i] * w for i in range(n))`
Confirms $\sum_i z_i e^{u_i}$; the F prefactor is folded into `charge_rhs`.

4. **Bulk Dirichlet BC** (l. 445–447).

`c0_i = max(float(c0_model[i]), _C_FLOOR)` then `fd.Constant(np.log(c0_i))`.
Confirms $u_i = \ln c_{0,i}$ on Γ_{conc} with floor $c_{0,i} \leftarrow \max(c_{0,i}, 10^{-20})$ for product species.

5. **Initial conditions** (l. 511–513). Same floor convention as the bulk BC; assigns $u_i^{(0)} = \ln c_{0,i}$.

6. **No blob IC**. The kwarg `set_initial_conditions(..., blob=True)` is silently ignored in log-c (see `dispatch.py:32–41`). The blob IC was a concentration-formulation feature with no log-c analog.

7. **η -clip on η_{scaled} , not on $\alpha n_e \eta_{\text{scaled}}$** (l. 244).

`min_value(max_value(eta_scaled, -clip_val), clip_val)`

Applied *before* multiplying by αn_e . This is the source of the $V_{\text{RHE}} > +0.495$ V R_2 unclip threshold (*not* the +1.14 V number from older handoffs).

8. **Log-rate BV uses unclamped u_i** (l. 337). The cathodic log-rate is built as

`log_cathodic = fd.ln(k0_j) + ui[cat_idx]`
`- alpha_j * n_e_j * eta_j`

Compare to the legacy branch (l. 369) which uses `c_surf[cat_idx] = exp(clamp(ui, +/-30))`.

7 What this transition did and did not change

Did change.

- Primary unknown $c_i \rightarrow u_i = \ln c_i$ on the dynamic species.
- Bulk Dirichlet BC type $c_i = c_{0,i} \rightarrow u_i = \ln c_{0,i}$; $c_{0,i} = 0$ now requires a floor.
- Time stepping is backward Euler in $c = e^u$, so the time term is $(e^{u^{n+1}} - e^{u^n})/\Delta t$, not $e^{u^{n+1}}(u^{n+1} - u^n)/\Delta t$.
- Concentrations are positive by construction, so the negative- c Newton iterates and the associated floor regularizations in the old solver are gone.
- Conditioning of Debye-layer regions is improved: the exponential in $c(x)$ becomes linear in $u(x)$.
- BV residual gains an algebraically rearranged log-rate option (`bv_log_rate=True`) that lets the surface u_i enter unclamped.

Did not change.

- The Poisson PDE structure (only its source uses e^{u_i}).
- The BV reaction rate physics, E_{eq} values, α , k_0 , stoichiometry, or the meaning of η .
- The η -clip ± 50 on η_{scaled} , which is floating-point safety on $\exp(\alpha n_e \eta)$ and is structural, not a c-vs-u choice.

- The bulk u -clamp ± 30 , which is the necessary log-c analog of guarding $\exp(u)$ as a coefficient and *cannot* be eliminated by the log-rate trick (different role: coefficient vs single deferred exp).
- The Boltzmann counterion (`add_boltzmann_counterion_residual`) is added orthogonally to the log-c transform.

8 Where to read more

- `docs/PNP Equation Formulations.tex` — original concentration formulation.
- `Forward/bv_solver/forms_logc.py` — production implementation.
- `docs/clipping_conventions.md` — detailed accounting of the three clips/clamps and why each one stays.
- `docs/bv_solver_unified_api.md` — how to call the production stack via config flags.